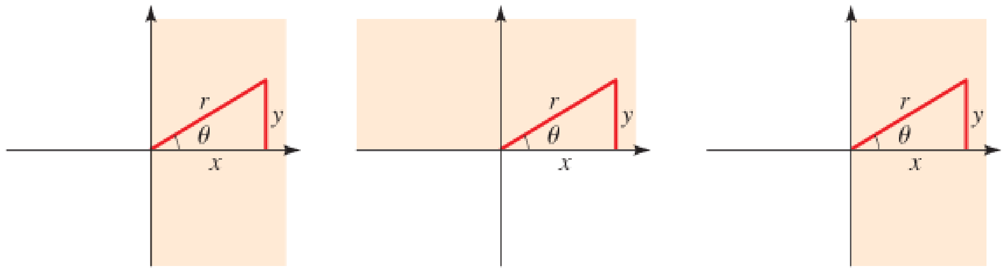
Ms. Welles **Alg 2 Trig/Honors**

Chp 5.4: Inverse Trigonometric Functions and Right Triangles

I. The Inverse Sine, Inverse Cosine, and Inverse Tangent Functions



$\sin\theta = \frac{y}{r}$ $\qquad$ $\cos\theta = \frac{x}{r}$ $\qquad$ $\tan\theta = \frac{y}{x}$

$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ $\qquad$ $0 \le \theta \le \pi$ $\qquad$ $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

The sine function, on the restricted domain $[-\pi/2, \pi/2]$; the cosine function, on the restricted domain $[0, \pi]$; and the tangent function, on the restricted domain $(-\pi/2, \pi/2)$; are all one-to-one and so have inverses.

The **inverse sine function** is the function $\sin^{-1}$ with domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$ defined by $\sin^{-1} x = y \Leftrightarrow \sin y = x$. The **inverse cosine function** is the function $\cos^{-1}$ with domain $[-1, 1]$ and range $[0, \pi]$ defined by $\cos^{-1} x = y \Leftrightarrow \cos y = x$. The **inverse tangent function** is the function $\tan^{-1}$ with domain R and range $(-\pi/2, \pi/2)$ defined by $\tan^{-1} x = y \Leftrightarrow \tan y = x$.

Function	Domain	Range
$\sin^{-1}$	[-1,1]	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}$	[-1,1]	$[0, \pi]$
$\tan^{-1}$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

The inverse sine function is also called arcsine, denoted by arcsin. The inverse cosine function is also called arccosine, denoted by arccos. The inverse tangent function is also called arctangent, denoted by arctan.

NOTE:

$$\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right) \quad (x \ge 1)$$

$$\csc^{-1} x = \sin^{-1}\left(\frac{1}{x}\right) \quad (x \ge 1)$$

$$\cot^{-1} x = \tan^{-1}\left(\frac{1}{x}\right) \quad (x > 0)$$

II. Solving for Angles in Right Triangles

The inverse trigonometric functions can be used to solve for angles in a right triangle if the lengths of at least two sides are known.

III. Evaluating Expressions Involving Inverse Trigonometric Functions

Expressions like $\sin\left(\tan^{-1}\frac{12}{5}\right)$ arise in calculus. We find exact values of such expressions using trigonometric identities or right triangles.

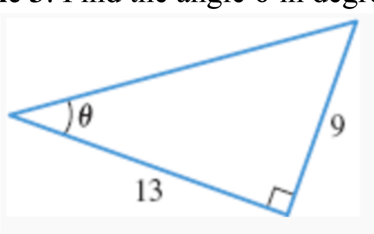
Example 1: Find the exact value of each expression, if it is defined. Express your answer in radians.

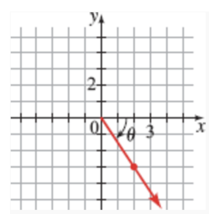
a. $sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ b. $tan^{-1}(-1)$ c. $sin^{-1}0$ d. $cos^{-1}(-1)$ e. $tan^{-1}\sqrt{3}$

Example 2: Use a calculator to find an approximate value of each expression rounded to three decimal places, if it is defined. State your answer in radians and degrees.

a. $cos^{-1}(-0.2)$ b. $sec^{-1}(5)$ c. $tan^{-1}3$ d. $sin^{-1}(-2)$

Example 3: Find the angle θ in degrees, rounded to one decimal place.

a. 

b. 

Example 4: Find all angles θ between 0° and 180^{0} that satisfies the given equation. Round your answer(s) to one decimal place.

a. $sin\,\theta = \frac{2}{3}$ b. $tan\,\theta =- 20$ c. $cos\ \theta =- \frac{2}{5}$

Example 5: Find the exact value of the expression.

a. $cos(tan^{-1}\left(\frac{4}{3}\right))$ b. $csc\left(cos^{-1}\left(\frac{-7}{25}\right)\right)$

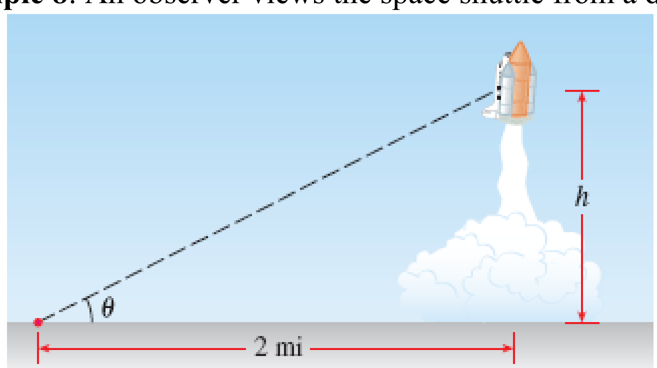
c. $sin\left(sec^{-1}(-4)\right)$ d. $cos\left(sin^{-1}\left(\frac{4}{5}\right)\right)$

Example 6: Rewrite the expression as an algebraic expression in *x*.

a. $cot(cos^{-1}x)$ b. $sec\left(tan^{-1}x\right)$

Example 7: A 96-foot tree casts a shadow that is 120 ft long. What is the angle of elevation of the sun?

Example 8: An observer views the space shuttle from a distance of 2 mi from the launch pad.



a. Express the height of the space shuttle as a function of the angle of elevation θ.

b. Express the angle of elevation θ as a function of the height *h* of the space shuttle.